

Report on Pattern Identification of Critical Combustion Events

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Abstract

This report presents methodologies for pattern identification in critical combustion events. The work focuses on the development and application of advanced data analysis techniques to understand and predict combustion instabilities and critical phenomena in reactive flow systems.

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1 Introduction

Understanding critical combustion events is a complex task that involves extensive analysis of both experimental and numerical data. Finding patterns in these datasets is an important step towards building better models for combustion phenomena. Many tools exist both in the reactive and non reactive flows. Proper Orthogonal Decomposition (POD) (CITAZIONE), is an Singular Value Decomposition (SVD) based method that provide a low rank representation of the flow rewriting it as a sum of orthogonal spatial patterns. Principal component analysis has been successfully employed to project high-dimensional data onto a lower-dimensional orthogonal subspace of thermodynamical variables (CITAZIONE), effectively reducing the complexity of the system while retaining its essential features. However, this method is not able to capture the temporal evolution of the system and it is not able to provide a physical interpretation of the modes. Dynamic Mode Decomposition (DMD) (CITAZIONE) allows the reduction of the system in spatio-temporal modes which have both a spatial structure and an associated temporal frequency and growth rate. Thus the system is represented as the sum of modes evolving with a linear dynamics in time that allows the study of the spectral properties of the system. Although powerful this method suffers from noisy data and it often is inadequate to provide an accurate description of turbulent reactive flows. High Order Dynamic Mode Decomposition (HODMD) (CITAZIONE) is an extension of DMD that improves the method by introducing the higher order Koopman hypothesis. This makes it more robust and has already been successfully applied in reactive flows (CITAZIONE). Combustion data often presents itself in tensorial form. Moreover the most valuable datasets, i.e. the ones that most closely represent the real physics of the system, are often the ones with the highest dimensionality both in spatial component and in the number of variables. This has sparked interest in methods that process data directly into this form and don't require reshaping into matrix form as in the case of SVD based methods and DMD.

The current report focuses on describing the application of these techniques to enhance the algorithms that allows pattern identification in combustion data. High Order Singular Value Decomposition (CITAZIONE), a generalization of SVD, is discussed as a tool to both compress and extract physical patterns from huge combustion datasets. In particular, this method allows for multi modal analysis along each axis of the tensor at the same time, allowing to quantify the importance of each dimension of the tensor. When combined with HODMD this, allows for identification of key feature in the dynamics of the reactive flow.

Furthermore,

eseguendo una riduzione della dimensionalità indipendente lungo ciascun modo del tensore — dimensioni spaziali, temporali e delle variabili — consentendo migliori tassi di compressione e un filtraggio del rumore più mirato. Combinata con l'HODMD, l'HOSVD fornisce un framework multi-dimensionale naturale per analizzare le dinamiche complesse e multi-scala dei flussi reattivi. Per arricchire ulteriormente l'interpretazione fisica di queste decomposizioni, la Decomposizione di Koopman Spazio-Temporale (STKD) [Le Clainche et al., 2020] estende l'HODMD al dominio spaziale, identificando pattern di onde viaggianti caratterizzati sia da frequenze che da numeri d'onda. Ciò consente di rappresentare il flusso come sovrapposizione di strutture propaganti, offrendo una comprensione più profonda dell'interazione tra dinamiche spaziali e temporali nella combustione turbolenta. Al di là degli approcci basati sulla decomposizione, modelli surrogati data-driven

come la Regressione con Processi Gaussiani (GPR) [Rasmussen Williams, 2006] offrono una prospettiva complementare, consentendo la ricostruzione probabilistica e la predizione dei campi di flusso a partire da osservazioni sparse. Quando estesa con la riduzione della dimensionalità basata su HOSVD, la GPR può operare in uno spazio tensoriale compresso, rendendola scalabile agli dataset ad alta dimensionalità generati da DNS e LES, preservando al contempo le capacità di quantificazione dell’incertezza. In questo lavoro, applichiamo ed estendiamo queste metodologie — HOSVD, HODMD, GPR potenziata con HOSVD e STKD — a dataset di combustione DNS e LES, con l’obiettivo di fornire un framework efficiente, fisicamente interpretabile e computazionalmente trattabile per l’analisi della dinamica dei flussi reattivi.

2 Datasets

The methodologies are assessed on different datasets from the consortium. These data describe high fidelity simulations of complex reacting flows.

2.1 NH_3/H_2 non-premixed mixing layer

This dataset consists of direct numerical simulations (DNS) of a two-dimensional non-premixed ammonia-hydrogen-air reacting mixing layer. It is generated by DCGIOVANNI at CORIA using the structured mesh reactive flow solver SiTCom-B [?]. The configuration represents a splitter-plate stabilized flame in which a cold fuel stream and a hot air stream interact across a shear layer, representative of conditions encountered in practical burners.

The lower stream consists of a NH_3/H_2 fuel blend (with 15% H_2 by volume) at ambient temperature (300 K) with a mean velocity of 15 m s^{-1} , while the upper stream consists of preheated air with a mean velocity of 39.6 m s^{-1} . A cold wall maintained at 600 K is imposed at the top boundary, and a symmetry condition is applied at the bottom. Homogeneous velocity fluctuations of 10% are prescribed at both inlets. The computational domain measures $12 \times 1 \text{ cm}$ and is discretized on a uniform grid of 2400×200 cells with a resolution of $50 \text{ }\mu\text{m}$.

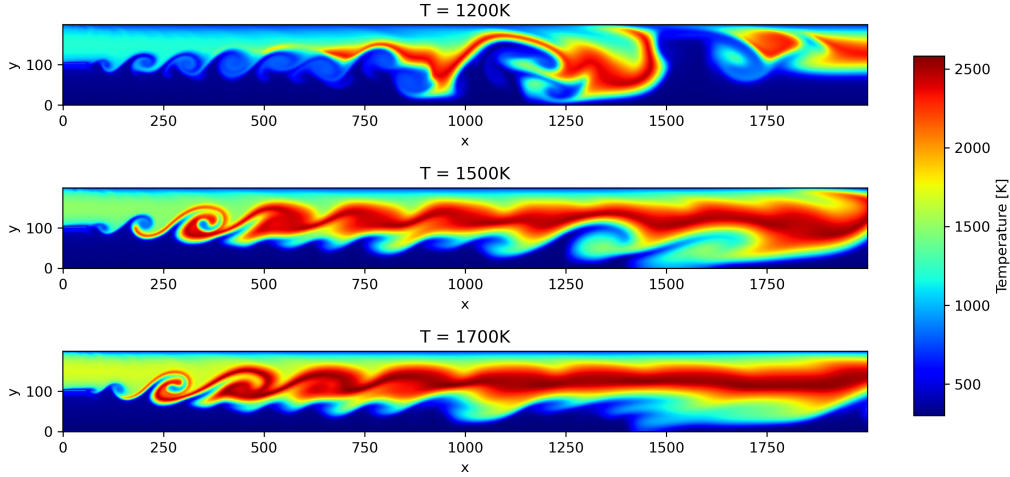


Figure 1: Three representative snapshots of the temperature field from the NH_3/H_2 non-premixed reacting mixing layer DNS, illustrating the spatial evolution of the flame and mixing structures at different instants.

Three datasets are generated at different air preheat temperatures, namely 1200 K, 1500 K, and 1700 K, to probe the sensitivity of the flame dynamics and species evolution to the inlet thermal boundary condition. The domain geometry, grid resolution, and flow velocities are identical across all three cases. Each snapshot contains 19 reactive scalar fields corresponding to the species of the reduced mechanisms RmAH-1 and RmAH-2 [?] — namely H_2 , O_2 , H_2O , H , O , OH , HO_2 , NH_3 , NH_2 , NH , N , NO , NO_2 , N_2O , HNO , NNH , N_2H_2 , N_2 , and Ar — together with temperature, yielding 20 variables in total. The species analysis presented in this work is conducted exclusively on the 1500 K dataset, which constitutes the reference operating condition for the assessment of the proposed methodologies.

References

- [1] Giovanni Grassi, Luc Vervisch, and Pascale Domingo. Reduced chemistry for numerical combustion of nh_3/h_2 fuel blend. *Combustion and Flame*, 279:114287, 2025.
- [2] Xavier Petit, Guillaume Ribert, G Lartigue, and P Domingo. Large-eddy simulation of supercritical fluid injection. *The Journal of Supercritical Fluids*, 84:61–73, 2013.
- [3] TODO. C3mechv3.5: Detailed chemical mechanism for ammonia-hydrogen combustion. *TODO*, 2023. Placeholder – replace with actual reference.